



MPCT: A medical image fusion method based on multi-scale pyramid convolution and Transformer

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ABSTRACT

In recent years, CNN and Transformer have been extensively used in medical image fusion. However, CNN is inherently limited in its ability to capture global and long-range semantic information due to the localized nature of convolutional operations. Meanwhile, the Transformer has limitations in learning local information and computational complexity. In addition, the existing fusion methods have a high information retention rate in one of the images and a low information retention rate in the other image when fusing two source images. To solve these problems, we propose a medical image fusion method (MPCT) based on multi-scale pyramid convolution and Transformer in this paper. Firstly, we design a multi-scale pyramidal convolutional network for extracting feature information at different levels. In addition, we design the Splice-CNN-Transformer (SCT) module to enhance the learning capabilities of global and long-range semantic information, optimizing the extracted features. We propose a similarity loss function that generates adaptive weights by calculating the similarity between the source and fused images, ensuring that the fused image maximally leverages information from both source images. Finally, with the comparative methods, our method has the least number of parameters. We also have extended the proposed methodology to address additional medical image fusion challenges. Experimental validations on public datasets indicate that our method outperforms other methods in both visual quality and quantitative results. Extension experiments also showcase the robust generalization capabilities of our method. The code of the method is available at <https://github.com/wangzi487794504/Image-fusion>.

1. Introduction

Due to the limitations of imaging principles, different medical imaging modalities can only capture part of the body structure and tissue information. Based on imaging information, medical images can be categorized into anatomical and functional images [1]. Anatomical images have high spatial resolution and can clearly show the anatomical structure of organs. For example, CT (Computed Tomography) images can well reflect dense structures such as bones and implants, and MRI (Magnetic Resonance Imaging) images can provide relevant information on soft tissues. However, anatomical images cannot show the functional changes of the human body. However, anatomical images do not show the functional changes in the body's metabolism. Instead, functional images can reflect functional and metabolic information, such as PET (Positron Emission Tomography) and SPECT (Single Photon Emission Computed Tomography). PET mainly reflects tumor function and metabolism, while SPECT images reflect tissue

organs and blood flow. However, due to the low resolution of functional images, anatomical details cannot be depicted [2]. In summary, the limitations of unimodal medical images have led to an increasing interest in multimodal medical image fusion. Multimodal medical image fusion aims to integrate important and complementary information from multiple images of different types into a single information-rich fusion image to help doctors diagnose diseases quickly and accurately, which can be used for clinical diagnosis, surgical navigation, and treatment planning [3].

To obtain information from the source image and extract various features from the source image for fusion. Many traditional methods have proposed various decomposition ways. These methods can be roughly divided into the following categories: methods based on sparse representation [4–6], methods based on multiscale transformations [7–10] [11], methods based on saliency [12], methods based on shallow

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machine learning [13–15], and methods based on the pulse coupled neural network (PCNN) [16–18]. However, their performance is highly dependent on the designed fusion rules, and to pursue better fusion performance, the design of fusion rules becomes more and more complicated, which makes it difficult to adapt to complex fusion scenarios. The rise of deep learning (DL) has provided more possibilities for image fusion. DL-based methods can be further categorized into four types based on network architecture: methods based on CNN [19,20], methods based on AE [21,22], methods based on GAN [23,24], and methods based on Transformer [25–28]. The basic components of the first three categories of methods are the convolutional layers. Consequently, they can only interact within the receptive field and cannot efficiently utilize global and remote semantic information to improve fusion results. Transformer-based methods use self-attention mechanisms to capture global and remote semantic information and have shown good performance in visual problems. But Transformer may not be as effective as CNN in processing local information, which focuses more on global information. The Transformer has higher computational complexity, which requires more computational resources and time for training and inference. In addition, the loss function used by most methods does not consider the importance of different source images in fusion. Thus, there is a problem that the information of one source image is sufficiently fused, but the information of another source image is not adequately fused.

To address the above problems, We propose a fusion method based on a multi-scale pyramid convolution and Transformer called MPCT. In this method, firstly, the network is designed in a multi-scale pyramid way, and the branches of different scales acquire features with different granularity by changing the size of the sensory field, which solves the problem of insufficient feature extraction. In addition, a Splice-CNN-Transformer (SCT) module is designed in the network to further enhance the learning capability of global and remote semantic information and optimize the feature information. It is composed of CNN and Transformer branches in parallel. Specifically, the CNN branch can adaptively adjust the weights of features in different regions and different channels of the image according to the feature information of the image space and channels. The Transformer branch effectively captures long-range dependencies between elements in the sequence, enhancing the learning ability of global and distant semantic information. Finally, we propose a similarity loss in the loss function by calculating the cosine similarity between the fusion feature map and the source image information to determine the information retention degree. A higher information retention degree corresponds to more preserved information in the source image, resulting in higher loss weight. When the two similarity values are closer, the loss function value is minimized. Conversely, when the discrepancy between the information retention of the fused image and the two source images is large, the loss function value increases, which prompts the network to learn from another source image to reduce the loss function value, thereby retaining the information from both source images. In summary, the main contributions of this work are as follows:

- (a) We propose the design of multi-scale pyramidal convolution in the network structure, where branches of different scales extract information of different granularity from low to high levels.
- (b) We design a Splice-CNN-Transformer (SCT) module that combines the respective strengths of CNN and Transformer to improve the model's ability to learn global and remote semantic information and optimize feature information.
- (c) To make the fused image fully retain the information of the two source images, we drive the network to learn the information of the two source images by calculating the similarity of the feature maps learned by the network to the source images and deriving two adaptive weights as the information retention degree. This approach addresses the limitations of most image fusion methods that struggle to adequately merge information from both source images or exhibit a bias towards a single source image.

- (d) Among the compared methods, our proposed MPCT method has the least number of parameters. We also extend the proposed method to handle other medical functional and structural image fusion tasks with good results. The experimental results verify that the model has good generalization ability. On mainstream datasets, the proposed MPCT outperforms other representative and state-of-the-art methods in terms of visual quality and quantitative results overall.

The remaining sections of this paper are organized as follows: Section 2 provides a review of related work and motivations of our work, including a brief overview of existing methods for multimodal medical image fusion, an introduction to Transformers, and the motivation behind this work. Section 3 presents a detailed explanation of the proposed method. Section 4 gives the experimental implementation details, fusion results, generalization to other medical image fusion, ablation analysis results, and comparison of parameters. Finally, Section 5 concludes the paper with a summary.

2. Related work

CNN and Transformer are the two most relevant technologies to this method. This paper reviews some representative research advancements.

2.1. Convolution-based method

With the successful application of deep learning in CV, DL-based research has become an active topic in the field of multimodal medical image fusion.

Among the CNN-based methods, Liu et al. [29] first propose the use of CNNs for multimodal medical image fusion tasks. They employ a convolutional network to generate a weight map that is used to integrate pixel activity information from two source images. MSR-PAN [30] proposes a multiscale residual pyramid attention network for medical image fusion and introduces a feature energy ratio strategy for the feature fusion process. Xu and Ma [31] use an image quality evaluation network and entropy to measure the image quality and the abundance of the source image to constrain the information extracted from the source image. Xu et al. [32] go further by proposing surface and deep constraints. To avoid manual tuning of hyper-parameters, Zhang and Ma [33] further propose squeezed decomposition networks and adaptive decision blocks to improve the fusion performance. Tang et al. [34] introduce semantic constraints in image fusion modeling and propose the gradient residual thick block, which improves the ability to describe fine-grained details. Jie et al. [35] introduce a joint target- and semantic-awareness-driven fusion network called TSJNet, which utilizes object- and semantically-relevant information gained from dual high-level tasks to guide the fusion network. Chang et al. [36] propose an independently trained multiscale alignment network based on image pyramids, which enables the full integration of image information at each scale.

The AE-based approach employs pre-trained autoencoders on a large-scale dataset as feature extractors and image reconstructors. Li et al. propose DenseFuse [37] as a pioneer in this fusion framework. Addressing previous CNN fusion methods where only the final layer's output is utilized as image features, causing significant feature decay, DenseFuse introduces dense blocks within the encoder to facilitate the flow of information and gradients between network layers, thereby enhancing ease of training. Li et al. [38] propose a novel Residual Fusion Network (RFN) to replace manual fusion strategies. Li et al. [22] a new cross-attention mechanism (CAM) to enhance complementary information. In addition, a fusion scheme based on a two-stage training strategy was proposed to generate fused images. Jian et al. [19] introduce feature interaction blocks to enhance the correlation between source and fused images by sharing each other's features. Wang et al. [39]

introduced a cross-attention module and a multi-stage fusion method, and the network can effectively extract and fuse features from different modalities to fully express the diversity of images. Negin Ghamsarian et al. [40] design the Deformable Pyramid Reception (DPR) module for performing shape-adaptive feature extraction on an input convolutional feature map.

The image fusion framework based on GAN effectively simulates data distribution in an unsupervised manner. Ma et al. define the image fusion problem as a game between the generator and the discriminator. They apply GAN to a series of fusion tasks. For instance, FusionGAN [41], used for the fusion of infrared and visible light images, marks the first instance of GAN adoption in handling image fusion tasks. Manisha Das et al. [42] use the content-aware generative adversarial network-based approach to improve the retention of feature information for each source image. Additionally, Li et al. [24] propose a Dual Attention Mechanism Based Nested Training Network (DANT-GAN), which utilizes two attention mechanisms for feature extraction and fusion of local regions and whole images, respectively. Fang et al. [23] propose an unsupervised two-channel dense convolution method DCD for multifocus image fusion.

2.2. Transformer-based method

The Transformer is proposed by Vaswani et al. [43] in 2017, incorporating the Attention Mechanism to form its model architecture. Following significant success in Natural Language Processing (NLP), the Transformer emerges as a major player in the field of computer vision. At present, Transformers are introduced into various visual tasks such as object detection [44], tracking [45], segmentation [46,47], and image restoration [48].

Due to its powerful long-range modeling capabilities, the Transformer is also introduced into the field of image fusion [49]. Building upon the AE-based fusion framework, Vibashan VS et al. [25] introduce IFT based on RFN-Nest, which incorporates an ST fusion block comprising a CNN and a Transformer branch to capture local and long-range features, respectively. Li et al. [50] propose a Convolution Guided Transformer Fusion (CGTF), aiming to combine the local features of CNNs and the long-range dependency features of Transformers to generate more satisfactory fusion results. Rao et al. [51] also introduce the Transformer into a GAN-based fusion framework, enabling the fusion of visible light and infrared images. Zhang et al. [52] design a branch network that integrates Transformer for interaction across multiple scales of fusion. This method extracts multi-scale complementary features from source images, thereby more effectively utilizing different forms of features compared to shared networks and parallel multi-branch structures. To effectively capture and utilize global contextual information from source images, [26] introduce adaptive convolution and adaptive transformers to address the multimodal medical image fusion problem. Ma et al. [49] propose a method that utilizes a combination of CNN and Swin Transformer to extract features containing both local and global information. These features are then fused within-domain and across-domains for the fusion process. Additionally, Zhao et al. [53] combine Transformers with CNNs to fully leverage the advantages of local context extraction and efficient computation. They introduce Invertible Neural Networks (INN) blocks to mutually generate input and output features, preventing information loss and addressing the issue of losing high-frequency information. Tang et al. [54] design Functional Guided Transformer Module (FGTM) and Anatomical Guided Transformer Module (AGTM) to communicate and collect perceptual features for further global complementary information interaction and aggregation. Wang et al. [55] design an L1-norm based on a sequence matrix and measure the corresponding activity level from the row and column vector dimensions that can well preserve the brightness and clear details of competitive images.

2.3. Motivations

In summary, methods based on convolutional neural networks rely on the powerful feature extraction and data fitting capabilities of neural networks to generate excellent fusion results. Although these convolutional neural network-based algorithms do show strong performance in image fusion tasks, they are inherently limited by fixed receptive fields. This constraint hinders their ability to establish long-distance complementary dependencies, resulting in a certain degree of blurring in the fused image.

The pure Transformer architecture lacks the advantages of CNNs in terms of translation invariance and locality and typically requires large amounts of training data or additional supervision to effectively perform computer vision tasks.

To better combine the strengths of CNNs with the advantages of Transformers, we propose a novel approach called multi-scale pyramid convolution and Transformer (MPCT). Recent methods have also explored the simultaneous processing of global and local features through CNN and Transformer branches. However, our approach offers three key advantages over these methods: Firstly, most methods do not fully utilize the information from both source images, often relying solely on the strengths of Transformers. In contrast, our method calculates the similarity between the source images and the fused image to generate adaptive weights, ensuring that the fused image maximizes the use of information from both source images. Secondly, many of methods typically input the source images into a single network without considering multi-scale information, which may lead to the loss of important details. We address this by incorporating a multi-scale pyramid convolutional design within the network architecture, where different branches extract information at varying levels from low to high scales. Lastly, our method is lightweight, with a minimal number of parameters, and employs a parallel structure that not only leverages the benefits of Transformers but also integrates the localization advantages of CNNs. This design effectively minimizes the potential negative impact of Transformers on CNN performance.

3. Proposed method

The network structure of the proposed multimodal medical image fusion method in this section is shown in Fig. 1. We use the symbol I_{SPECT} for the color source image and I_{MRI} for the grayscale source image. Considering that I_{SPECT} is an RGB image and I_{MRI} is a grayscale image, RGB-to-YUV color conversion is used to solve the channel mismatch and color loss problem. First, I_{SPECT} is converted to the YUV color space to obtain the Y, U, and V components: I_{SPECT}^Y , I_{SPECT}^U and I_{SPECT}^V . Then, I_{SPECT}^Y and I_{MRI} are concatenated in the channel dimension and sent to the proposed MPCT network. The Y component of the fused image is generated after network learning, and then the final fused image I_F is obtained by YUV-to-RGB color conversion between I_{SPECT}^U and I_{SPECT}^V .

3.1. Multi-scale pyramid convolution

Existing deep learning-based methods usually input source images into a single neural network, failing to fully utilize information between different scales. This may lead to the loss of critical information. In designing the network, we introduce the concept of the multi-scale pyramid. On the one hand, we employ the multi-scale method to incorporate varying numbers of convolution kernels within the same image size. These kernels are employed across different scales to extract information ranging from low to high levels. On the other hand, we draw inspiration from the feature pyramid network, utilizing downsampling operations to decrease the size of the feature map and broaden the receptive field.

Specifically, we define a basic feature extraction module (BM): it consists of a convolutional layer (Conv), a batch normalization (BN),

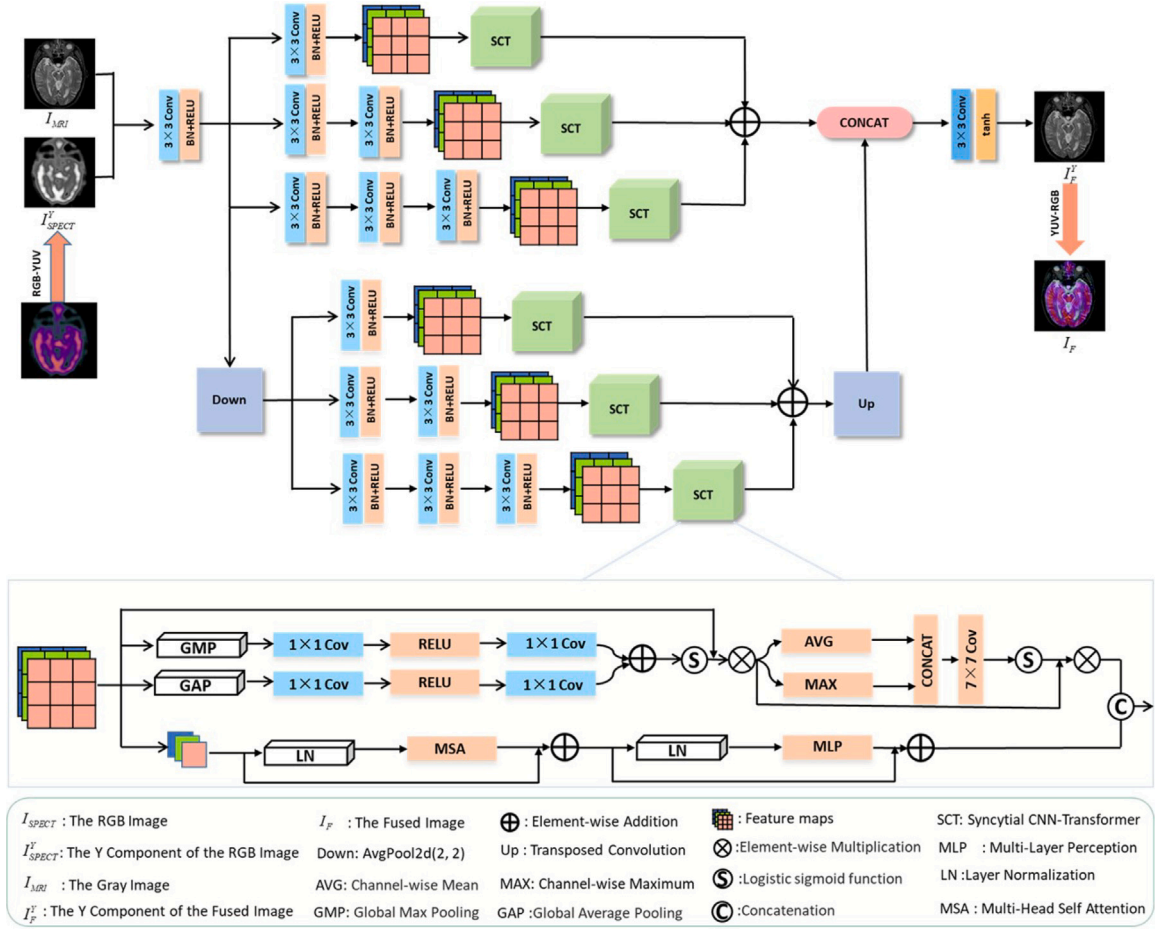


Fig. 1. The comprehensive framework of the proposed MPCT network and a detailed elucidation of its SCT module structure. The symbol ‘+’ denotes the addition operation performed across channels corresponding to different branches. ‘S’ is the abbreviation for the Sigmoid activation function, ‘X’ represents the multiplication operation performed on channels corresponding to different branches, and ‘C’ refers to concatenating elements from different branches along the channel dimension.

and a nonlinear activation function (RELU). Splicing I_{SPBCT}^Y and I_{MRI} in the channel dimension and inputting them into a BM, the above operation can be expressed as in Eq. (1):

$$F_{BM}^{Out} = RELU(BN(Conv(I_{in}))) \quad (1)$$

where F_{BM}^{Out} denotes the output features after passing through a BM module. $RELU$ is the activation function. BN is the normalization process. $Conv$ is the convolution operation. I_{in} denotes the input image features.

With the same image size, the three branches are arranged in a multi-scale manner to extract different levels of information from low to high levels. Among the three branches, the top branch features one BM module and one SCT module, the middle branch includes two BM modules and one SCT module, and the bottom branch comprises three BM modules and one SCT module. Three feature maps, denoted as φ_{C1} , φ_{C2} , and φ_{C3} , can be obtained from these branches. The more branches of the BM module can be extracted at a deeper level, thus allowing the extraction of fine-grained features. This structure facilitates the generation of fusion results with richer information. Finally, the inter-scale features are then feature pooled to improve the local and global learning ability of the network. The multiscale results can be expressed as Eq. (2):

$$\begin{aligned} F_{in} &= CAT(I_{Spect}^Y, I_{MRI}) \\ \varphi_{C1} &= F_{SCT}(F_{BM}^{Out}(F_{BM}^{Out}(F_{in}))) \\ \varphi_{C2} &= F_{SCT}(F_{BM}^{Out}(F_{BM}^{Out}(F_{BM}^{Out}(F_{in})))) \\ \varphi_{C3} &= F_{SCT}(F_{BM}^{Out}(F_{BM}^{Out}(F_{BM}^{Out}(F_{BM}^{Out}(F_{in})))) \end{aligned} \quad (2)$$

where φ_{C1} , φ_{C2} , and φ_{C3} represent the outputs of the three branches at the first level of the pyramid. F_{SCT} denotes the SCT module in the network. CAT signifies concatenation across channel dimensions. I_{Spect}^Y stands for the Y-component of a color image. I_{MRI} represents a grayscale image.

In addition, drawing on the idea of feature pyramid networks, we perform a downsampling operation on the input features as the second layer of input to the pyramid. Through downsampling, the range of neuron perception can be expanded, and a larger receptive field can better capture the overall features of the input data. However, downsampling is essentially information compression, which can lead to some information loss. In order to weigh the relationship between the expansion of the perceptual field and the loss of information, it is found after ablation experiments that the fusion effect reaches its best when the pyramid has two layers. Therefore, after downsampling, we obtained three output feature maps: φ_{D1} , φ_{D2} , and φ_{D3} . This process can be represented as Eq. (3):

$$\begin{aligned} \varphi_{D1} &= F_{Up}(F_{SCT}(F_{BM}^{Out}(F_{BM}^{Out}(F_{Down}(F_{in})))))) \\ \varphi_{D2} &= F_{Up}(F_{SCT}(F_{BM}^{Out}(F_{BM}^{Out}(F_{BM}^{Out}(F_{Down}(F_{in})))))) \\ \varphi_{D3} &= F_{Up}(F_{SCT}(F_{BM}^{Out}(F_{BM}^{Out}(F_{BM}^{Out}(F_{BM}^{Out}(F_{Down}(F_{in})))))) \end{aligned} \quad (3)$$

where φ_{D1} , φ_{D2} , φ_{D3} denote the outputs of the three branches at the second level of the pyramid. F_{Down} represents downsampling operations. F_{Up} denotes upsampling operations.

Finally, the outputs of the three branches in the first layer are combined with the outputs of the three branches in the second layer to aggregate the features extracted from each path. These aggregated features are then sent through a convolutional layer and, ultimately,

through the input of the T_{tanh} activation function to obtain the final fusion result denoted as I_F^V . I_F^V can be represented as Eq. (4):

$$I_F^V = \text{Tanh}(\text{Conv}(\text{CAT}\{\varphi_{C1}, \varphi_{C2}, \varphi_{C3}, \varphi_{D1}, \varphi_{D2}, \varphi_{D3}\})) \quad (4)$$

The overall description of the proposed MPCT is summarized in 1.

3.2. SCT module

In order to capture global and remote semantic information and optimize feature information, we designed the SCT fusion module by combining the advantages of CNN's local context extraction and computational efficiency with the Transformer's global focus and remote dependency modeling to accomplish the task. The architecture of the SCT module is shown in Fig. 1.

The SCT is composed of branches from both CNNs and Transformers. The CNN branch is designed to extract local information by allocating weights based on importance across channel dimensions and spatial regions. The Transformer branch optimizes feature information on a global semantic level. The CNN branch and the global Transformer branch are connected through SCT, using a parallel structure to link the optimized features with global semantic features. This approach minimizes the impact of the Transformer on the CNN, effectively integrating the two types of features.

The specific process involves first extracting the mean and maximum values from each channel through a global average pooling operation. Subsequently, a channel attention vector is generated via a fully connected layer. This vector is then normalized using a Sigmoid activation function and applied to each channel's feature map to ascertain the importance of different channels. The output of the channel weighting process yields F_{SCT}^S . Channel attention maps are then generated from the feature maps within each channel. Spatial information among the features is subsequently obtained by convolutions to produce the final spatial attention map, which is then applied to the input feature maps to derive the CNN branching output denoted as F_{SCT}^C , enhancing the model's ability to focus on spatial locations. F_{SCT}^C can be represented as Eq. (5):

$$\begin{aligned} F_{SCT}^S &= \text{Mul}(\text{Sig}(\text{Conv}(\text{RELU}(\text{GMP}(\varphi_{SCT}^{In}))) + \text{Conv} \\ &\quad (\text{RELU}(\text{GAP}(\varphi_{SCT}^{In}))), \varphi_{SCT}^{In})) \\ F_{SCT}^C &= \text{Mul}(\text{Sig}(\text{Conv}(\text{CAT}(\text{AVG}(F_{SCT}^S(\varphi_{SCT}^{In})), \text{MAX} \\ &\quad (F_{SCT}^S(\varphi_{SCT}^{In})))), F_{SCT}^S(\varphi_{SCT}^{In})) \end{aligned} \quad (5)$$

where Mul is the multiplication operation. Sig is the Sigmoid activation function. GMP is the adaptive maximum pooling. GAP is the adaptive average pooling. φ_{SCT}^{In} is the input to the SCT module. AVG and MAX are the operations for finding the mean and the maximum. F_{SCT}^S is the output of the CNN branch channel weighting. F_{SCT}^C is the output of the CNN branch.

On the Transformer branch, its output feature F_{SCT}^G expression is Eq. (6):

$$F_{SCT}^G = \text{MLP}(\text{LN}(\text{MSA}(\text{LN}(\varphi_{SCT}^{In}) + \varphi_{SCT}^{In})) + \text{MSA} \\ ((\text{LN}(\varphi_{SCT}^{In}) + \varphi_{SCT}^{In})) \quad (6)$$

where LN denotes layer normalization. MSA denotes multinomial self-attention. $+$ operation is denoted as residual connection.

Finally, the output result F_{SCT}^{Out} of the SCT module is the concatenation of the CNN branch and the Transformer branch. the SCT module output F_{SCT}^{Out} can be represented as Eq. (7):

$$F_{SCT}^{Out} = \text{CAT}(F_{SCT}^G(\varphi_{SCT}^{In}), F_{SCT}^C(\varphi_{SCT}^{In})) \quad (7)$$

where F_{SCT}^{Out} is the output of the SCT module.

Algorithm 1: Overall description of MPCT

Input: I_a : The MRI image, I_b^* : The (PET/SPECT/CT) image
Output: I_f^* : The Fused image

```

1 begin
2   Initialize the MPCT network;
3   foreach Iteration rounds  $epoch = 1, 2, \dots, 90$  do
4     if Channel of  $I_b^* == 3$  then
5       into YUV color space to get  $I_b$ , the Y component of
6          $I_b^*$ 
7     end
8     Normalize  $I_a$  and  $I_b$ ;
9     Combine  $I_a$  and  $I_b$ , input them into MPCT Net, and
10      obtain the fused feature map  $I_f$ ;
11     Calculate the similarity between the feature map and
12      the source image by cosine similarity;
13     Compute the similarity weights of the feature map and
14      the source image by Eq. (12);
15     Update the parameters of MPCT Net by minimizing
16       $L_{total}$  defined in Eq. (8);
17     if Channel of  $I_b^* == 3$  then
18       Merge  $I_f$  with the U and V components of  $I_b^*$  and
19       then apply inverse YUV transform to get the
20       normalized pseudo-color fused image,  $I_f^*$ ;
21     else
22        $I_f^* = I_f$ ;
23     end
24     obtain the fused image,  $I_f^*$ ;
25   end
26 end

```

3.3. Loss function

For SPECT/PET images, color information represents various functional organizations. Utilizing this color information helps locate the location of lesions. Therefore, it is essential to preserve color information as much as possible to maintain the intensity distribution of the Y channel in the fused image. For MRI images, they consist of white high signals and black low signals. Many lesions exhibit signals stronger than the surrounding normal tissues, often appearing as bright regions. Hence, preserving the white texture information is more valuable as it captures significant details in the image. Additionally, based on attenuation values, CT images can be divided into darker low-density areas (soft tissue) and brighter high-density areas (bone), providing detailed anatomical information of the skeletal structure for image fusion. Based on the above analysis, image features primarily consist of two forms: pixel intensity and texture information. Therefore, the fused image has to maintain significant pixel intensity to preserve dense structural and functional information. In addition, rich textures should be preserved to retain the organizational information. In summary, the network loss function expression is Eq. (8):

$$L_{total} = L_{tex} + L_{int} \quad (8)$$

where L_{total} is the total loss function of the MPCT network. L_{tex} is the image texture loss. L_{int} is the image intensity loss.

For the network to extract texture information from the image while ensuring minimal distortion of the overall image, we use the Structural Similarity Index Measure (SSIM) to separately constrain the image in three aspects: correlation, luminance, and contrast distortions. This constraint aims to reduce information loss and distortion during the fusion process, and we refer to this loss as global texture loss. In addition, although SSIM can constrain the image to be distortion-free, it does not take into account the effect of the corresponding pixel points in the two source images on the fusion result, which is

a probabilistic issue. Cross-entropy measures the accuracy of model learning by comparing the probability distribution of the model output with the source image output. Therefore, adopting the cross-entropy loss function reduces the difference between the fused image and the source images, better ensuring the preservation of image texture. We refer to this as detail texture loss. The expression for texture loss is as follows Eq. (9):

$$L_{texl} = \alpha L_g + \beta L_{en} \quad (9)$$

where L_g is the global texture loss of the image. L_{en} is the detailed texture loss of the image. α is the hyperparameter of L_g . β is the hyperparameter of L_{en} .

The global texture loss L_g is calculated as Eq. (10):

$$L_g = \omega_1 * (1 - SSIM(I_f, I_1)) + \omega_2 * (1 - SSIM(I_f, I_2)) \quad (10)$$

where $SSIM(\cdot)$ denotes the structural similarity operation. I_1 and I_2 stand for source images. I_f is the fused image. ω_1, ω_2 are the importance weights of the source image I_1 and source image I_2 respectively.

The detail texture loss L_{en} is given by Eq. (11):

$$L_{en} = \omega_1 * L_{bce}(I_f, I_1) + \omega_2 * L_{bce}(I_f, I_2) \quad (11)$$

$$L_{bce} = -(I_f * \log(Sig(I_{in})) + (1 - I_f) * \log(1 - Sig(I_{in})))$$

where Sig is the activation function that maps I_{in} to the range (0, 1). $\log$ denotes the natural logarithm with e as the base.

Past image fusion methods can fully fuse the information of one of the source images. However, the fusion of information from another source image is relatively insufficient. To fully obtain the information of the two source images, we give two adaptive weights as the information preservation degree, which we call L_{sim} . The corresponding weights are derived by calculating the similarity between the feature maps learned by the network and the source images. The higher the similarity, the higher the weights, the larger the corresponding loss value of the source images, and the higher the information preservation degree of the corresponding source images. The loss function achieves the minimum value only when the similarity between the fused image and the two source images is close. The two adaptive weights are denoted as ω_1 and ω_2 respectively, with the calculation formula Eq. (12):

$$[\omega_1, \omega_2] = \text{softmax}([V_1, V_2])$$

$$V_i = \max\left(\frac{I_f I_{in}}{\|I_f\| \|I_{in}\|}, 0\right) \quad (12)$$

where I_f denotes the fused image and I_{in} denotes the input image. V_1 and V_2 are the similarities computed between the two source images and the fused image. These similarities are mapped to the range of 0–1 using the softmax function, ensuring that the weights of ω_1 and ω_2 sum up to 1.

Although L_g and L_{en} can constrain the intensity distribution, the texture loss primarily focuses on contrast and structural changes rather than the differences in intensity distribution. To better constrain the intensity of each pixel, we adopt the mean squared error (MSE) as the loss function. MSE can calculate the difference between each fused pixel and the source image. By minimizing the MSE, we can ensure that the intensity of the fused image closely matches the intensity of the source image. This approach helps the fused image retain more information from the source image, thereby enhancing overall image quality. However, since the MSE is an average of the results calculated for each pixel point, the result obtained may have a blurred appearance. We use L_g to structurally constrain the image, which can reduce the blurriness of the fused image and compensate for this issue. Therefore, our intensity loss L_{int} is calculated using the following formula Eq. (13):

$$L_{int} = \gamma * (\omega_1 * MSE(I_f, I_1) + \omega_2 * MSE(I_f, I_2)) \quad (13)$$

where γ is the intensity loss hyperparameter.

4. Experiments

In this section, we first give the dataset and training details. Subsequently, we compare MPCT with several state-of-the-art algorithms on fused images through qualitative results and quantitative results. Extended experiments on other visual tasks also demonstrate the good generalization ability of our approach. Finally, we compare the number of parameters of deep learning-based methods and validate the effectiveness of a particular design through a series of ablation studies.

4.1. Datasets and training details

The dataset for medical image fusion is downloaded from the publicly available Harvard Medical Image Dataset.¹ It is worth mentioning that Harvard is a widely used dataset for multi-modal medical image fusion [26,32,54]. We downloaded 357 pairs of SPECT and MRI images, each with dimensions of 256×256 pixels. These downloaded images are randomly divided into a training set, a validation set, and a test set. For CT and MRI image fusion, we downloaded 160 image pairs. Due to the limited number of CT training samples that might not significantly impact the network's performance, we implemented a two-stage training strategy. In the first stage, we trained the network using infrared and visible datasets and leveraged the M3FD dataset, which comprises 275 images cropped to 256×256 pixels. Subsequently, in the second stage, we conducted training using the CT dataset.

The experiments are implemented on a Tesla T4 GPU through the PyTorch framework. During training, the Adam optimizer is used with a learning rate of 0.001 and an Epoch of 90. For SPECT images, the batch size is set to 6, and hyperparameters are set to 15. For CT and MRI image fusion, the batch size is set to 4, and hyperparameters are set to 6.

4.2. Evaluation indicators

Combining the characteristics of medical images, we select ten metrics to evaluate the fusion performance: entropy (EN), structural similarity index measure (SSIM), correlation coefficient (CC), mean square error (MSE), peak signal-to-noise ratio (PSNR), total correlation Difference (SCD) and multi-scale structural similarity index measure (MS-SSIM). The main reasons are as follows:

EN [56] is a measure of how much information is contained in the image, and a higher EN means that the fused image is richer in information.

SSIM [57] is used to measure the structural similarity between the source image and the fused image, with larger SSIM values representing less image distortion.

CC [58] is mainly used to measure the linear correlation between two images, and a larger value of CC indicates that more structural and textural information about the image is preserved.

SCD [59] measures the correlation between the fused image and the source image by calculating the difference between them. A positive result suggests that the fused image effectively preserves certain information from the source image. In contrast, a negative result indicates that the fused image inadequately retains the information from the source image or even contradicts it.

MSE [60] is utilized to quantify the discrepancy between two images, reflecting the fusion outcome in comparison to a reference image at the pixel level. A smaller MSE indicates a closer resemblance between the fusion outcome and the reference image, signifying superior performance.

PSNR [61] characterizes the ratio of peak power to noise power in the fused image, reflecting the distortion in the fusion process at the

¹ <http://www.med.harvard.edu/AANLIB/home.html>

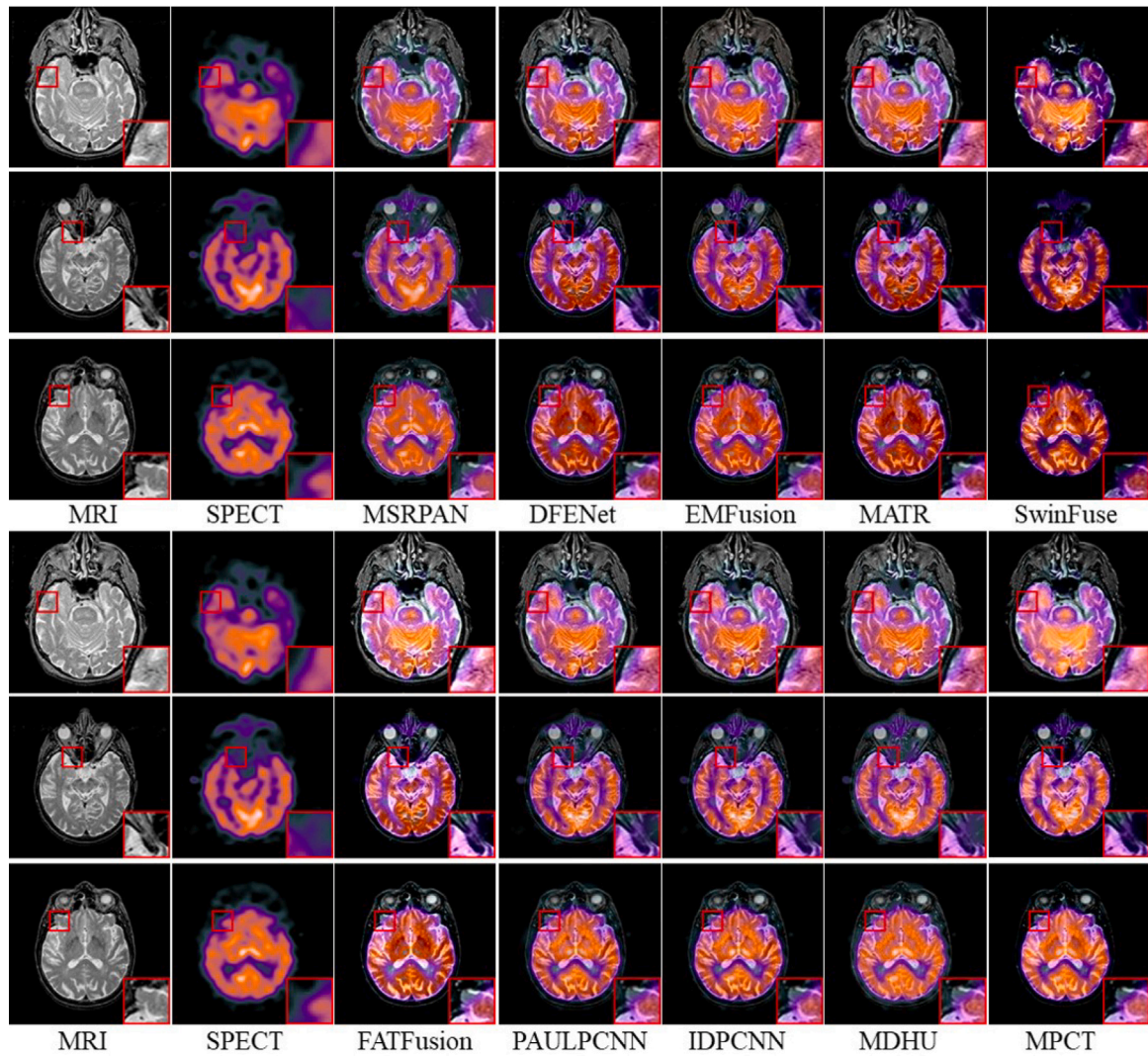


Fig. 2. Qualitative comparison of our MPCT with nine methods on three MRI and SPECT image pairs.

pixel level. A higher PSNR indicates that the distortion of the fused image is smaller and that the fused image is closer to the source image.

MS-SSIM [62] calculates the similarity based on the structural information and perceptual properties of an image, which is used to evaluate the quality of the image.

Q_{NICE} [63] measures the nonlinear correlation information entropy preserved in the fused image.

Q_{CV} [64] metric is used to quantify the visual quality of fused images. It extracts the quality change before and after fusion by comparing the similarity between the fused image and the reference image.

Q_{TE} [65] measures the dependence degree between the input and the fused images, which is a divergence measurement.

4.3. Results

We compare MPCT with nine state-of-the-art methods: A multiscale residual pyramid attention network for medical image fusion [30] (MSRPN), A dual-branch feature enhanced network integrating transformers and convolutional feature learning for multimodal medical image fusion [27] (DFENet), An unsupervised enhanced medical image fusion network [32] (EMFusion), Multimodal medical image fusion via multiscale adaptive transformer [26] (MATR), A residual swin transformer fusion network for infrared and visible images [55] (SwinFuse), A functional-anatomical transformer for medical image fusion [54]

(FATFusion), Parameter adaptive unit-linking pulse coupled neural network based MRI-PET/SPECT image fusion [17] (PAULPCNN), Multi-modal medical image fusion using improved dual-channel PCNN [18] (IDPCNN), Multi-modal medical image fusion via multi-dictionary and truncated Huber filtering [6] (MDHU). Qualitative results and quantitative results are provided in each fusion task.

4.3.1. SPECT and MRI image fusion

We apply this method to the fusion task of SPECT and MRI, presenting three sets of fusion results for SPECT and MRI pairs as shown in Fig. 2. The enlarged area in the figure shows that MSRPN cannot effectively extract the low-density areas of MRI. EMFusion and SwinFuse can extract the overall structural information of MRI, but they cannot constrain the luminance information well. For example, the high signal of the ventricle (white in the center area of the picture) of MRI becomes a low signal in gray color after fusion by EMFusion. MATR extracts the MRI more adequately but loses the brightness information from SPECT. FATFusion alters the brightness information from MRI. MDHU takes into account the high-density areas in SPECT and MRI, but artifacts appear in the low densities. Our method shows three advantages. Firstly, it can extract detailed textural information from MRI images, as evident from the first row where low signal areas in the MRI image are well fused. Secondly, in the fusion images of competitors, the emphasis is only on one source image's information. Our method can retain

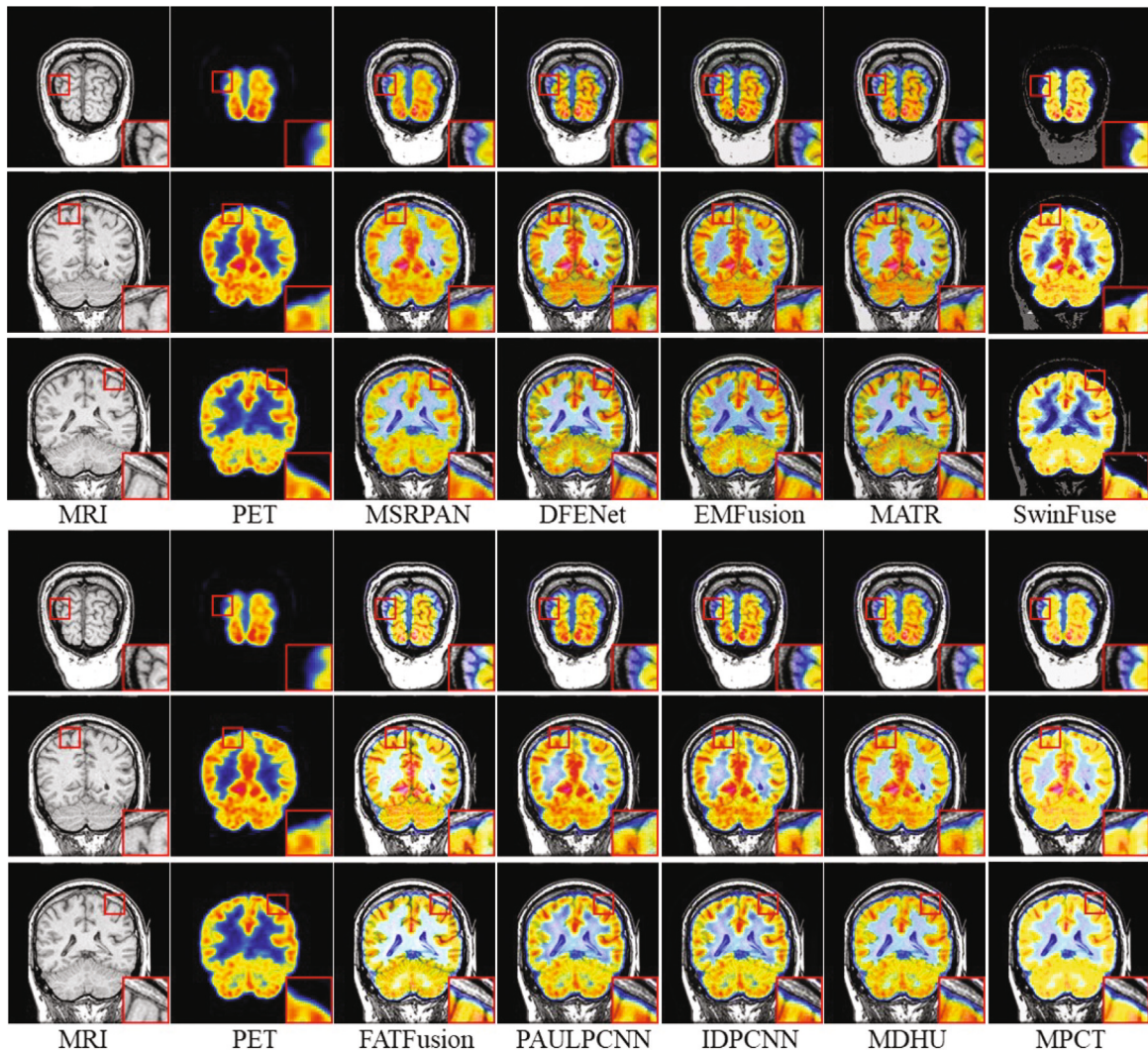


Fig. 3. Qualitative comparison of our MPCT with nine methods on three MRI and PET image pairs.

crucial information from both source images simultaneously. Thirdly, in competitor methods, the Y-component from SPECT is lost, whereas our method effectively preserves the Y-component of SPECT.

The quantitative results are shown in Table 1. Our method achieved the best results in EN, CC, SCD, and MS-SSIM, the second-best result in TE, and ranked third in SSIM and MSE. These quantitative results demonstrate that the fusion outcomes maintain a high level of structural similarity and information content with the two source images. Therefore, our results effectively preserve the structural information from MRI images and the functional information from SPECT images. Furthermore, information preservation in images has consistently been a bottleneck for small networks. By incorporating an image intensity loss function, we achieved favorable values in the EN and TE metrics, indicating that the network effectively retains image information.

4.3.2. PET and MRI image fusion

To investigate the generalization capability of this method on other types of multimodal medical images, we extend MPCT to the task of PET and MRI image fusion. We directly test the model trained on SPECT without adjustments. We conduct a generalization experiment using 24 pairs of PET and MRI test samples.

We provide the fusion results of three pairs of PET and MRI images as shown in Fig. 3. While all ten methods demonstrate strong fusion performance, some issues persist. Specifically, The MSRPAN method cannot fully retain the complementary features, resulting in blurred

fusion results. The EMFusion and MATR-based methods distort the color information of PET to some extent. While SwinFuse effectively extracts functional information from PET images, it compromises the brightness information from MRI, resulting in a fused image with a grayscale distribution. MDHU and PAULPCNN lost the dark color information in the PET section. Overall, our proposed MPCT demonstrates the best fusion performance in simultaneously utilizing the functional information of PET images and the tissue information of MRI images.

The quantitative results are shown in Table 2. CC, SCD, and Q_{NICE} achieve the best results among the ten methods. EN, MSE, PSNR, MS-SSIM, Q_{CV} , and Q_{TE} are all in the top three. The overall performance of the method is better than other algorithms by comparison of visual quality and objective evaluation, with better generalization ability.

4.3.3. CT and MRI image fusion

We apply the method to the CT and MRI fusion task and provide fusion results for three pairs of CT and MRI images, as shown in Fig. 4. All of these ten methods achieve more satisfactory fusion performance, but there are still some problems. In the first example, the MSRPAN method appears as a dentate shape at the edge of the magnified region of the image. Furthermore, the low-density region in the CT becomes larger in extent after fusion. In the fusion results of FATFusion and SwinFuse, the CT densities are diminished, the luminance information is lost, and multiple artifacts are generated, making the overall structure of the fused image slightly distorted. Compared with other methods, MPCT

Table 1

The quantitative comparison results between the proposed MPCT method and six competitors in SPECT and MRI image fusion are as follows for 24 test image pairs. The quantitative results of fusion outcomes across various metrics are displayed (Displaying average values, where red indicates the best, blue indicates the second-best, and green indicates the third-best).

Metrics	MSRPAN	DFENet	EMFusion	MATR	SwinFuse	FATFusion	PAULPCNN	IDPCNN	MDHU	MPCT
EN	4.7533	4.3920	4.9179	5.0486	3.8842	4.9281	5.2701	5.2947	4.6456	5.4384
SSIM	0.8863	0.9387	0.9543	0.9200	0.8296	0.9265	0.9713	0.9749	0.9448	0.9566
CC	0.8656	0.8412	0.8771	0.8409	0.8452	0.8456	0.8727	0.8713	0.8701	0.8900
SCD	0.9595	0.8055	0.9357	-0.010	0.4307	1.5293	1.2850	1.3051	1.2897	1.5475
MSE	0.0148	0.0173	0.0122	0.0165	0.0160	0.0265	0.0159	0.0161	0.017	0.0158
PSNR	66.698	65.994	67.509	66.124	66.293	64.146	66.371	66.332	66.052	66.361
MS-SSIM	0.9028	0.9036	0.9468	0.8890	0.8004	0.8890	0.9585	0.9607	0.9314	0.9636
Q_{NICE}	0.8116	0.8172	0.8079	0.8128	0.8074	0.8129	0.8076	0.8072	0.8141	0.8106
Q_{CV}	266.7	30.6	58.5	46.3	841.1	85.2	90.5	77.3	134.63	115.7
Q_{TE}	0.4623	0.4977	0.4696	0.7562	0.4172	0.4649	0.4803	0.4894	0.4760	0.7437

Table 2

The quantitative comparison results between the proposed MPCT method and nine competitors in PET and MRI image fusion are as follows for 24 test image pairs. The quantitative results of fusion outcomes across various metrics are displayed (Displaying average values, where red indicates the best, blue indicates the second-best, and green indicates the third-best).

Metrics	MSRPAN	DFENet	EMFusion	MATR	SwinFuse	FATFusion	PAULPCNN	IDPCNN	MDHU	MPCT
EN	5.0736	5.2273	5.6412	5.6665	3.7749	5.513	5.6052	6.0343	5.0749	5.9991
SSIM	0.8568	0.9159	0.935	0.8784	0.7054	0.8886	0.9615	0.9653	0.92219	0.92223
CC	0.7877	0.7678	0.8019	0.7704	0.8166	0.7771	0.8033	0.8044	0.7995	0.8349
SCD	1.2177	0.8448	1.2207	0.3001	1.244	1.3812	1.2695	1.2817	1.2842	1.6769
MSE	0.0511	0.0524	0.0415	0.0458	0.0448	0.0669	0.0519	0.0509	0.0566	0.0488
PSNR	61.131	61.016	62.041	61.58	61.674	59.936	61.096	61.141	60.989	61.33
MS-SSIM	0.9028	0.8392	0.8925	0.7962	0.7085	0.8135	0.9403	0.9402	0.8899	0.9142
Q_{NICE}	0.8111	0.8162	0.8083	0.8140	0.8065	0.8124	0.8089	0.8077	0.8144	0.8142
Q_{CV}	580.2	59.9	67.0	149.2	2785.9	131.1	92.3	94.8	75.3	119.1
Q_{TE}	0.4579	0.5366	0.4911	0.7683	0.4061	0.4758	0.4726	0.5225	0.4834	0.8779

Table 3

The quantitative comparison results between the proposed MPCT method and nine competitors in CT and MRI image fusion are as follows for 24 test image pairs. The quantitative results of fusion outcomes across various metrics are displayed (Displaying average values, where red indicates the best, blue indicates the second-best, and green indicates the third-best).

Metrics	MSRPAN	DFENet	EMFusion	MATR	SwinFuse	FATFusion	PAULPCNN	IDPCNN	MDHU	MPCT
EN	4.2022	3.9997	4.2249	4.8524	4.5663	4.5514	5.4738	5.3576	4.1851	5.4295
SSIM	0.8164	0.8635	0.7373	0.7831	0.4682	0.7854	0.8843	0.9435	0.8972	0.8653
CC	0.7674	0.7807	0.7157	0.6948	0.6058	0.7092	0.8026	0.7911	0.784	0.8013
SCD	1.3194	1.4973	0.7148	-0.1696	0.4334	0.9375	1.4708	1.5174	1.5111	1.3531
MSE	0.048	0.0499	0.047	0.0487	0.0632	0.055	0.0423	0.0438	0.0496	0.0380
PSNR	61.419	61.227	61.49	61.343	60.186	60.804	61.961	61.806	61.269	62.443
MS-SSIM	0.8519	0.8869	0.7161	0.5809	0.3379	0.5973	0.8576	0.94185	0.9022	0.9093
Q_{NICE}	0.8106	0.8110	0.8096	0.8122	0.8094	0.8105	0.8061	0.8061	0.8089	0.8124
Q_{CV}	2580.4	1484.8	890.1	7600.5	1643.7	8051.3	2519.7	966.3	1431.6	1407.8
Q_{TE}	0.4791	0.5252	0.5007	0.8619	0.4953	0.5006	0.5664	0.5732	0.4509	0.8388

preserves more texture details and luminance information. Our results effectively preserve the dense structure and luminance information of CT images.

In addition, our results retain more information in the MRI image without interfering with the CT image by calculating the similarity loss weights in the loss function. In the fusion results of EMFusion and MATR, it is found that some of the black low signals and white high signals in the magnified region of MRI are lost in the fusion results, which is not sufficient for the fusion of MRI image information. DFENet, PAULPCNN, IDPCNN, and MDHU lose part of the black low signal. The purpose of MRI and CT image fusion is to preserve the tissue information (bright areas in the figure, such as gray matter brain tissue, blood vessels, lesions, etc.) of MRI and the structural information (such as bone structure) of CT. In the fusion result of our method, it can be seen that the zoomed-in region has both the low-density information of CT and the high-signal tissue information of MRI. Therefore, information from both source images can be retained simultaneously in the fusion result of MPCT.

Table 3 reports the quantitative results for seven metrics. The proposed MPCT achieves optimal results in MSE, PSNR, and Q_{NICE} .

It achieves the second-best result in EN, CC, MS-SSIM, and Q_{TE} indicating that the fused image can contain rich source image information while achieving good similarity with the source image and efficiently preserving the texture and structure information of the source image.

4.4. Ablation study

4.4.1. Multi-scale pyramid layer ablation study

We investigate the impact of the number of layers in feature extraction through experiments to identify the optimal number of layers for the pyramid. As depicted in Fig. 5, panel Fig. 5(a) represents a single-layer multi-scale pyramid network, while panel Fig. 5(b) showcases a three-layer multi-scale pyramid network, both of which are tested on the trained model.

The quantitative results are presented in Table 4, where the first column indicates the name of the evaluation metrics, and the second, third, and fourth columns indicate the results of the tests using one, two, and three layers of the multiscale pyramid, respectively. The qualitative findings are illustrated in Fig. 6. The experimental results show that the two-layer network model produces the best results. Most

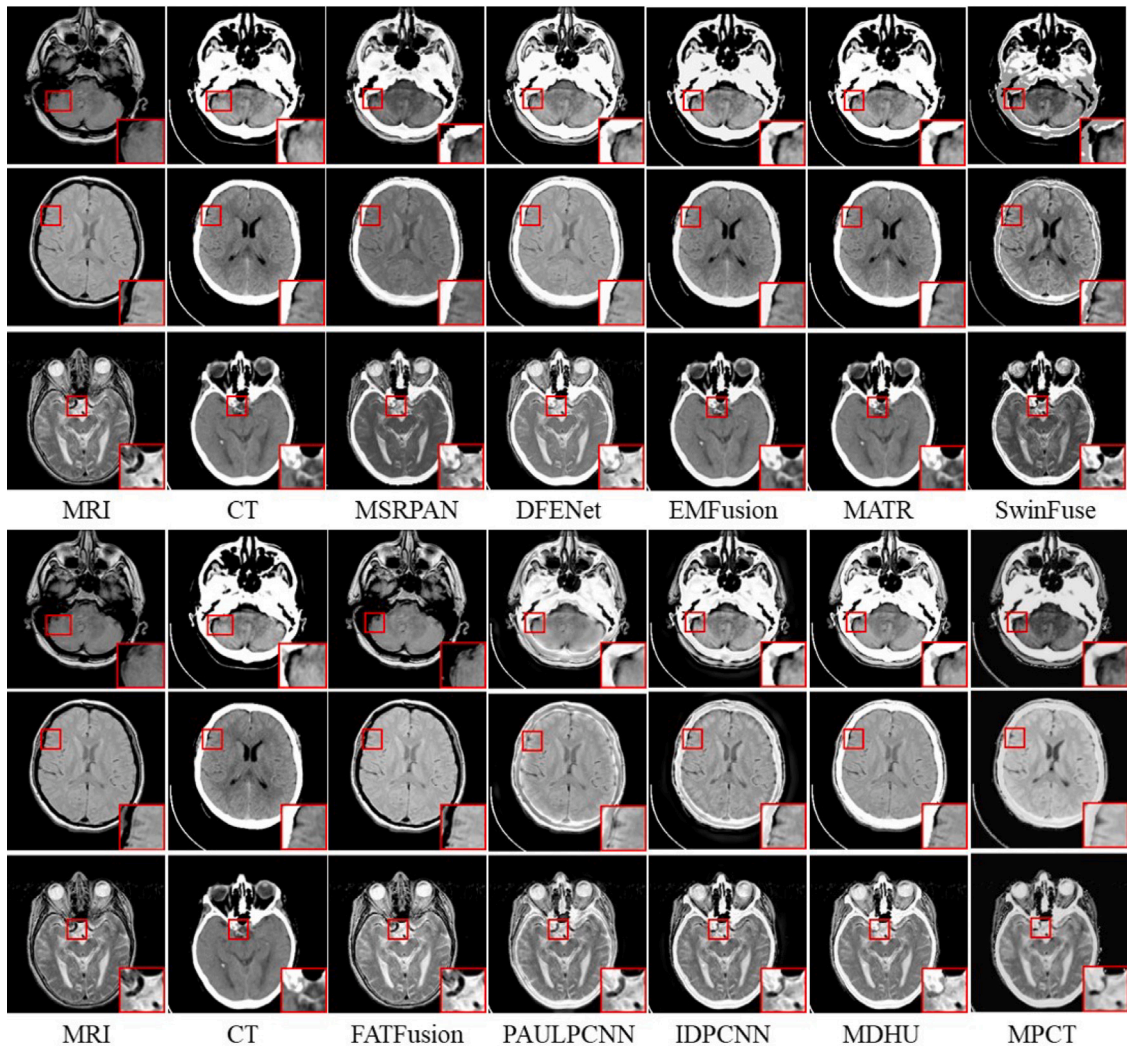


Fig. 4. Qualitative comparison of our MPCT with nine methods on three MRI and CT image pairs.

Table 4

Quantitative results of pyramid layer ablation studies (mean values shown, best values in red).

Metrics	SPECT and MRI image fusion			CT and MRI image fusion		
	First	Second	Third	First	Second	Third
EN	4.8426	5.4384	5.0195	4.7835	5.4295	4.9512
SSIM	0.8720	0.9566	0.9200	0.7879	0.8653	0.7367
CC	0.8711	0.8900	0.8906	0.6265	0.8013	0.6188
SCD	-0.2131	1.5475	-0.0339	0.0620	1.3531	-0.1055
MSE	0.0169	0.0158	0.0186	0.0520	0.0380	0.0463
PSNR	65.886	66.361	65.701	61.013	62.443	61.515
MS-SSIM	0.8519	0.9636	0.8890	0.7950	0.9093	0.7352
Q_{NICE}	0.8057	0.8106	0.8071	0.8025	0.8124	0.8066
Q_{CV}	705.6	115.7	160.9	1017.6	1407.8	1282.1

of the metrics showed significant improvement compared to the other two models.

4.4.2. SCT ablation analysis

To illustrate that SCT has the ability to learn and optimize feature information with global and remote semantic information, we conduct an ablation experimental study. The SCT is removed from the whole architecture, and the quantitative results of removing the SCT and the original network are given in Table 5. In Figs. 7 and 8, two sets of qualitative results for SPECT/MRI and CT/MRI are respectively provided. It is known that the complete model is overall better than the network model with the SCT module removed, and it can obtain better results than the degraded network in terms of metrics as a whole.

4.4.3. Analysis of similarity loss ablation

To investigate the influence of similarity on fusion outcomes, we conduct an ablation study on it. Specifically, we remove the similarity loss from the loss function to demonstrate its necessity.

In the quantitative results, the network with similarity loss outperforms the network with similarity loss removed in every metric, as shown in Tables 6, which demonstrates the effectiveness of similarity loss. In addition, on the two sets of SPECT/MRI in Fig. 9 and the two sets of CT/MRI image characterization results in Fig. 10, the network with similarity loss removed can only effectively extract SPECT and CT information and cannot extract the brightness information of MRI. In contrast, the use of similarity loss constraints enables the network to retain the two source image information.

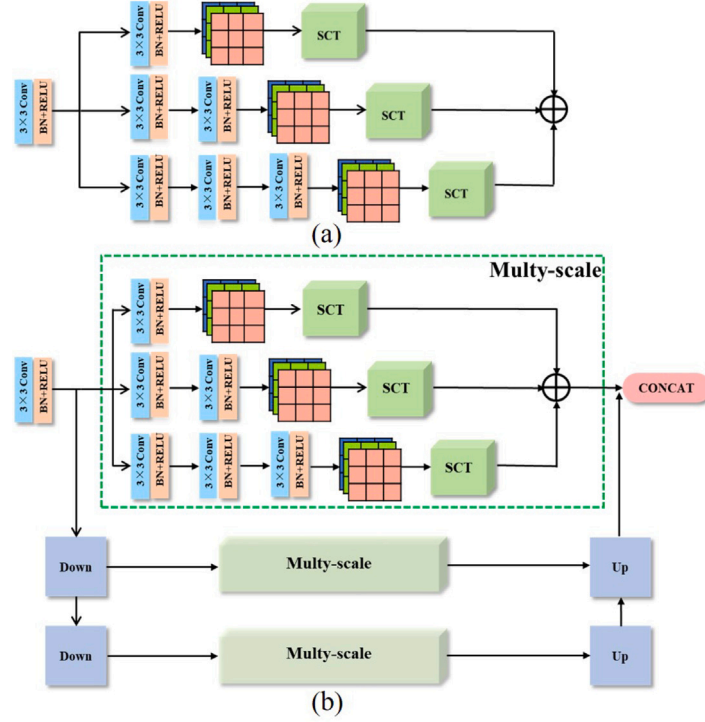


Fig. 5. Two different network structures for the proposed ablation study method. (a): One-layer multi-scale pyramid. (b): Three-layer multiscale pyramid.

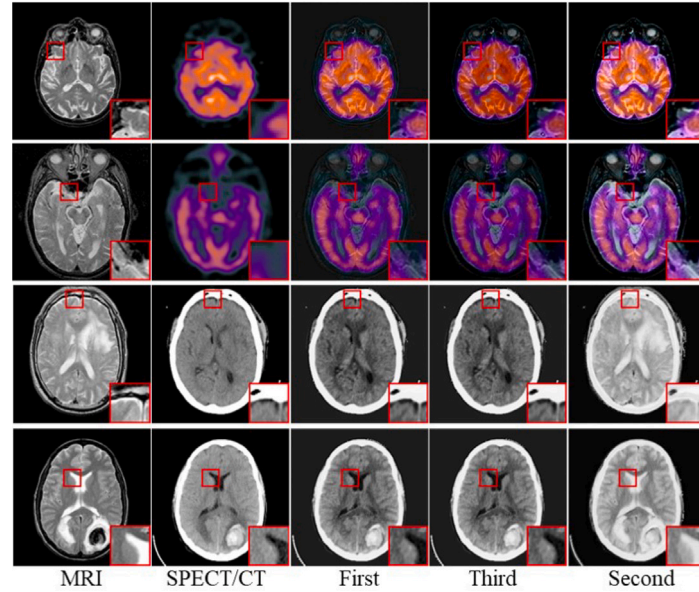


Fig. 6. Qualitative comparison of SPECT-MRI and CT-MRI image pairs to validate the effectiveness of the feature extractor at different pyramid layers. The first two rows show the results for MRI and SPECT images, and the last two rows show the results for MR and CT images.

4.4.4. Loss function erosion and parameter analysis

We conduct several experiments on the test set aimed at verifying the effectiveness of the loss function. Specifically, we remove texture losses $L_{W_{ithOut}}^g$ and $L_{W_{ithOut}}^n$, as well as intensity loss $L_{W_{ithOut}}^{int}$ from the loss function and evaluate the qualitative results of the subsequent network, as shown in Fig. 11. When L_g is removed, a distortion of the overall image is observed in the fusion results of the first and second rows of SPECT images, suggesting that L_g has a constraining effect on the image texture. When L_g is removed, a distortion of the overall image is observed in the fusion results of the first and second rows of SPECT images, suggesting that L_g has a constraining effect on

the image texture. Furthermore, in the fused images using $L_{W_{ithOut}}^n$, a loss of high signal in the MRI is observed in the zoomed-in region, suggesting a loss of detailed texture. In contrast, in the qualitative results of MPCT, the detailed texture information of MRI remains intact, highlighting the significant role of L_{en} in extracting detailed texture information. Finally, some transparent points appear at the edges of the fused images in case $L_{W_{ithOut}}^{int}$, which is not observed when utilizing Network L_{int} . This observation highlights the superiority of MPCT fusion over the ablation network. The conclusion is further supported by the quantitative results in Table 8, which show that MPCT outperforms the ablation network in all evaluation metrics.

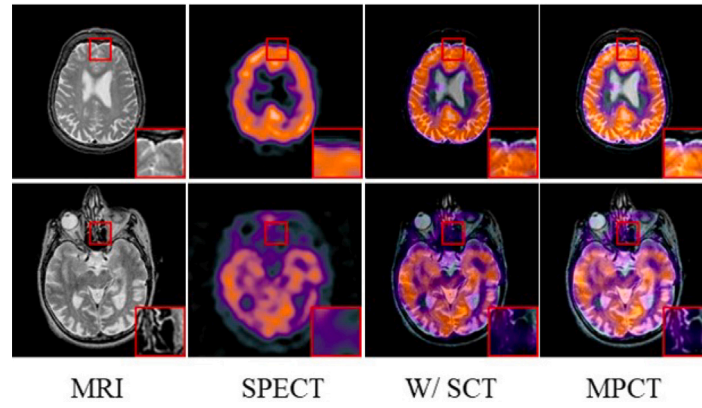


Fig. 7. Two sets of SPECT/MRI source images and their corresponding fusion results under removal of the SCT module. From left to right: MRI, SPECT, W/SCT (fusion results with SCT module removed).

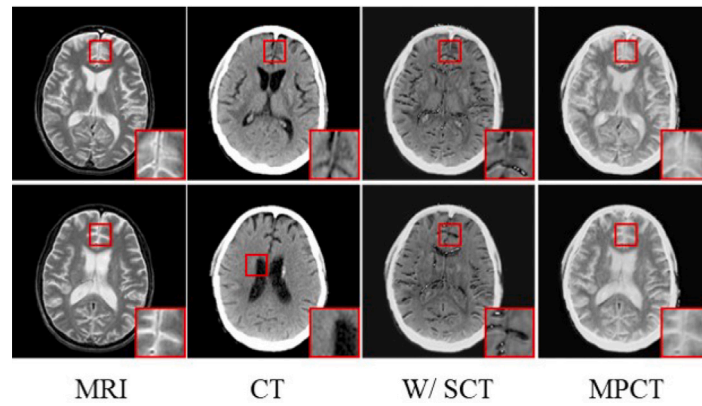


Fig. 8. Two sets of CT/MRI source images and their corresponding fusion results under removal of the SCT module. From left to right: MRI, CT, W/SCT (fusion results with SCT module removed).

Table 5

Quantitative results of the SCT ablation study on the CT/MRI dataset and on the SPECT/MRI dataset (mean values shown, red: best).

Metrics	SPECT and MRI image fusion		CT and MRI image fusion	
	Without SCT	MPCT	Without SCT	MPCT
EN	4.9913	5.4384	5.4324	5.4295
SSIM	0.5080	0.9566	0.7552	0.8653
CC	0.6816	0.8900	0.6788	0.8013
SCD	-0.1852	1.5475	0.3502	1.3531
MSE	0.0270	0.0158	0.0433	0.0380
PSNR	63.943	66.361	61.816	62.443
MS-SSIM	0.4378	0.9636	0.7375	0.9093
Q_{NICE}	0.8041	0.8106	0.8124	0.8039
Q_{CV}	2730.7	115.7	1488.2	1407.8
Q_{TE}	0.7267	0.7437	0.8382	0.8388

Table 6

Quantitative results of L_{sim} ablation studies on the CT/MRI dataset and on the SPECT/MRI dataset (mean values shown, red: best).

Metrics	SPECT and MRI image fusion		CT and MRI image fusion	
	Without L_{sim}	MPCT	Without L_{sim}	MPCT
EN	4.8619	5.4384	5.2609	5.4295
SSIM	0.8815	0.9566	0.8482	0.8653
CC	0.8060	0.8900	0.7533	0.8013
SCD	-0.1182	1.5475	0.7362	1.3531
MSE	0.0164	0.0158	0.0398	0.0380
PSNR	66.045	66.361	62.273	62.443
MS-SSIM	0.8873	0.9636	0.8866	0.9093
Q_{NICE}	0.8041	0.8106	0.8026	0.8124
Q_{CV}	2730.7	115.7	1488.2	1407.8
Q_{TE}	0.4782	0.7437	0.8318	0.8388

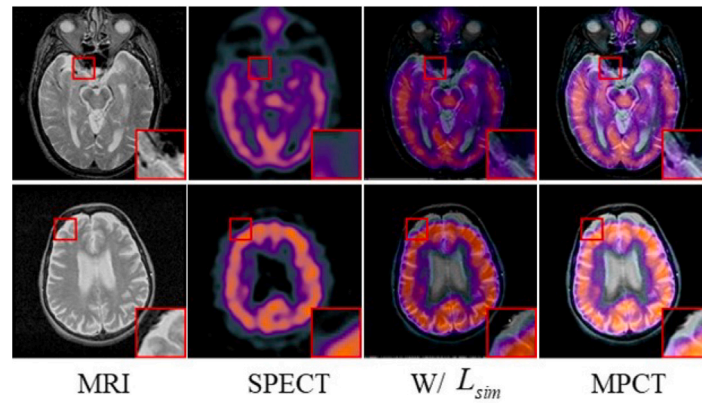


Fig. 9. Two sets of SPECT/MRI source images and their corresponding qualitative fusion results with similarity loss removed. From left to right: MRI, SPECT, W/ L_{sim} (L_{sim} removed), MPCT.

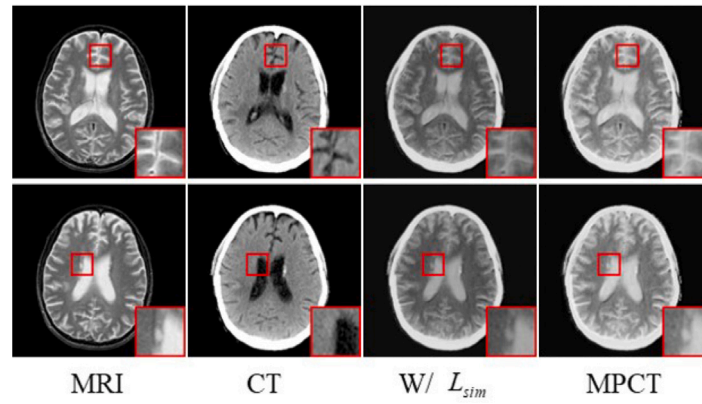


Fig. 10. Two sets of CT/MRI source images and their corresponding qualitative fusion results with similarity loss removed. From left to right: MRI, CT, W/ L_{sim} (L_{sim} removed), MPCT.

Table 7

Parameters of deep learning-based fusion methods (red: optimal).

Methods	MSRPAN	DFENet	FATFusion	SwinFuse	EMFusion	MATR	MPCT
Parameters	2 050 049	22 475 127	6 223 297	8 085 104	297 283	107 574	76185

The second objective is to explore the weighting parameters (α , β , and γ) within the loss function. Fig. 12 presents the parameter tuning of the network on the SPECT dataset, ultimately determining $\alpha = 15$, $\beta = 1$, and $\gamma = 1$. Due to the numerous combinations of these three hyperparameters, it is neither suitable nor necessary to display them all. Hence, a set of results is provided using a controlled variable method. Specifically, to demonstrate the impact of a particular weight parameter, the other two factors are set to default values. For each image in the test set, a quantitative evaluation is performed, where the x-axis represents the values of the hyperparameter, and the y-axis shows the average score across all samples in the test set. The fusion performance on the validation set is optimal when $\alpha = 15$, $\beta = 1$, and $\gamma = 1$. Consequently, we set the default values for α , β , and γ as 15, 1, and 1, respectively.

4.5. Parameter comparison

We compare the parameter sizes of deep learning-based fusion methods, including MSRPAN [30], DFENet [27], EMFusion [32], MATR [26], SwinFuse [55], FATFusion [54], and our proposed MPCT. We focus on networks used solely for testing. For instance, in autoencoder-based methods, only the generator is employed for fusion without the need for the discriminator during testing. Thus, we consider only the generator's parameters. For approaches using pre-trained

models to extract features and generate images based on specific rules during testing, we account for the pre-trained model's parameters. The parameter quantities for deep learning-based methods are shown in Table 7. Compared to others, our method has the fewest parameters.

5. Conclusion

In this paper, we introduce MPCT, a novel medical image fusion method based on multiscale pyramid convolution and Transformer. The MPCT method incorporates a multiscale pyramid convolutional network, extracting information across different levels from low to high scales. Additionally, we propose a Splice-CNN-Transformer (SCT) module to enhance global and long-range semantic information learning while optimizing the extracted features. To fully utilize information from both source images, a similarity loss in the loss function is introduced, employing computed similarity between the source and fused images to generate adaptive weights, preserving information from both sources. Finally, MPCT is trained in an unsupervised manner. Experimental results on a public dataset demonstrate that MPCT effectively preserves structural information and rich texture details of source images in the context of multimodal medical image fusion and outperforms other representative and state-of-the-art methods in terms of visual quality and objective evaluation as a whole. Our method can combine the medical information from two images, assisting doctors in

Table 8

Quantitative results of ablation studies in CT/MRI dataset and SPECT/MRI dataset with the removal of specific Loss (show mean, red: best).

Metrics	SPECT and MRI image fusion				CT and MRI image fusion			
	$L_{WithOut}^g$	$L_{WithOut}^{en}$	$L_{WithOut}^{int}$	MPCT	$L_{WithOut}^g$	$L_{WithOut}^{en}$	$L_{WithOut}^{int}$	MPCT
EN	2.8714	4.7501	4.8383	5.4384	4.5336	5.3101	5.1102	5.4295
SSIM	0.6860	0.8296	0.8125	0.9566	0.7062	0.8090	0.8286	0.8653
CC	0.6580	0.8865	0.7866	0.8900	0.7029	0.7438	0.7515	0.8013
SCD	0.4904	-0.4838	0.0724	1.5475	0.8481	1.1684	1.1982	1.3531
MSE	0.0330	0.0159	0.0177	0.0158	0.0502	0.0463	0.0413	0.0380
PSNR	63.079	66.252	65.788	66.361	61.185	61.567	62.093	62.443
MS-SSIM	0.6527	0.7864	0.8167	0.9636	0.7321	0.8255	0.8152	0.9093
Q_{NICE}	0.8046	0.8060	0.8065	0.8106	0.8051	0.8065	0.8070	0.8124
Q_{CV}	1293.3	592.5	679.9	115.7	2178.6	1099.0	928.5	1407.8
Q_{TE}	0.7257	0.7432	0.6987	0.7437	0.8386	0.6948	0.8248	0.8388

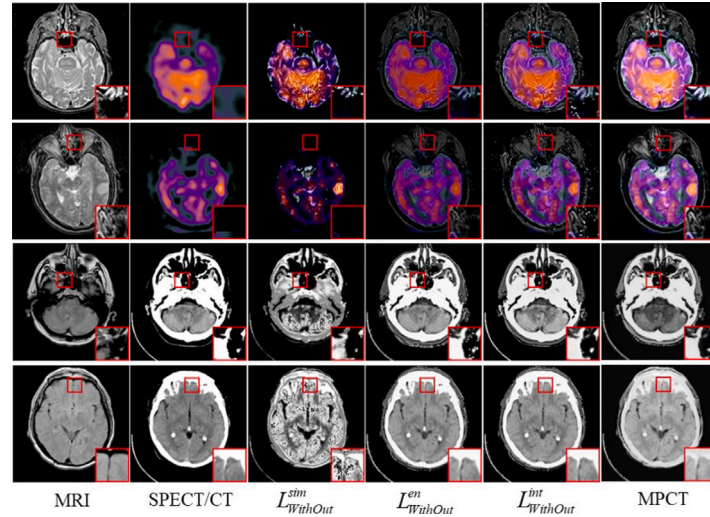


Fig. 11. Two sets of SPECT/MRI source images and two sets of CT/MRI source images, and their corresponding fusion results under removal of specific Loss. From left to right: MRI images, SPECT/CT, $L_{WithOut}^g$, $L_{WithOut}^{en}$, $L_{WithOut}^{int}$, MPCT.

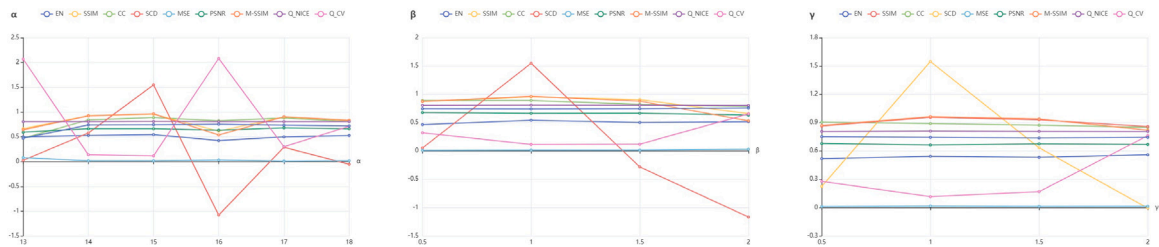


Fig. 12. Weighing the effects of the parameters α , β , and γ on the quantization performance. The horizontal coordinates are the values of each of the three hyperparameters, and the vertical coordinates are the average values in the test set. To show the variations of the data more clearly in the images, we shrink the PSNR by a factor of 100, the QCV by a factor of 1000 and the EN by a factor of 10.

quickly and accurately diagnosing brain diseases. Moreover, due to the reduced number of parameters, our network requires less time to generate the fused images. However, there are still some shortcomings. In future work, we will consider using color information to provide richer complementary information for the network, and more consideration will be given to the actual needs of subsequent visual tasks to enhance support for clinical medicine.

CRedit authorship contribution statement

Yi Xu: Supervision, Resources. **Zijie Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation. **ShouCai Wu:** Writing – review & editing. **XiongFei Zhan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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